



FORENSIC ENVELOPE AUDIT • V1.0

Strategic Thermal Reconnaissance Report

142 Bluegrass Ridge • Lexington, KY • Scan 2026-06-24 • Ground-truth accuracy ±0.78 cm

TOTAL SQUARES

24.31

SQ (100 sf)

SHINGLE LAYERS

2 / 2

DETECTED / CODE MAX

FACETS

11

DISTINCT PLANES

ANOMALIES FLAGGED

3

FORENSIC HOTSPOTS

EXECUTIVE SUMMARY • GRAND TOTAL RANGE

\$32,491.38 → \$33,996.38

MATERIALS \$23,407.50 • LABOR \$6,028.88 • TEAR-OFF \$1,850.00 • UNFORESEEN \$1,205.00-\$2,710.00

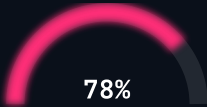
STRATEX RECOMMENDATION

TEAR OFF

Two existing shingle layers + 88% sub-surface moisture probability exceeds code re-roof limit. Full tear-off, decking remediation, and new-install is the only insurable path.

Building Envelope Score

ENERGY DEFICIENCY



DEFICIENT

VENTILATION



POOR

OVERALL ENVELOPE

55/100

COMPOSITE

AI-PRIORITIZED MAINTENANCE QUEUE

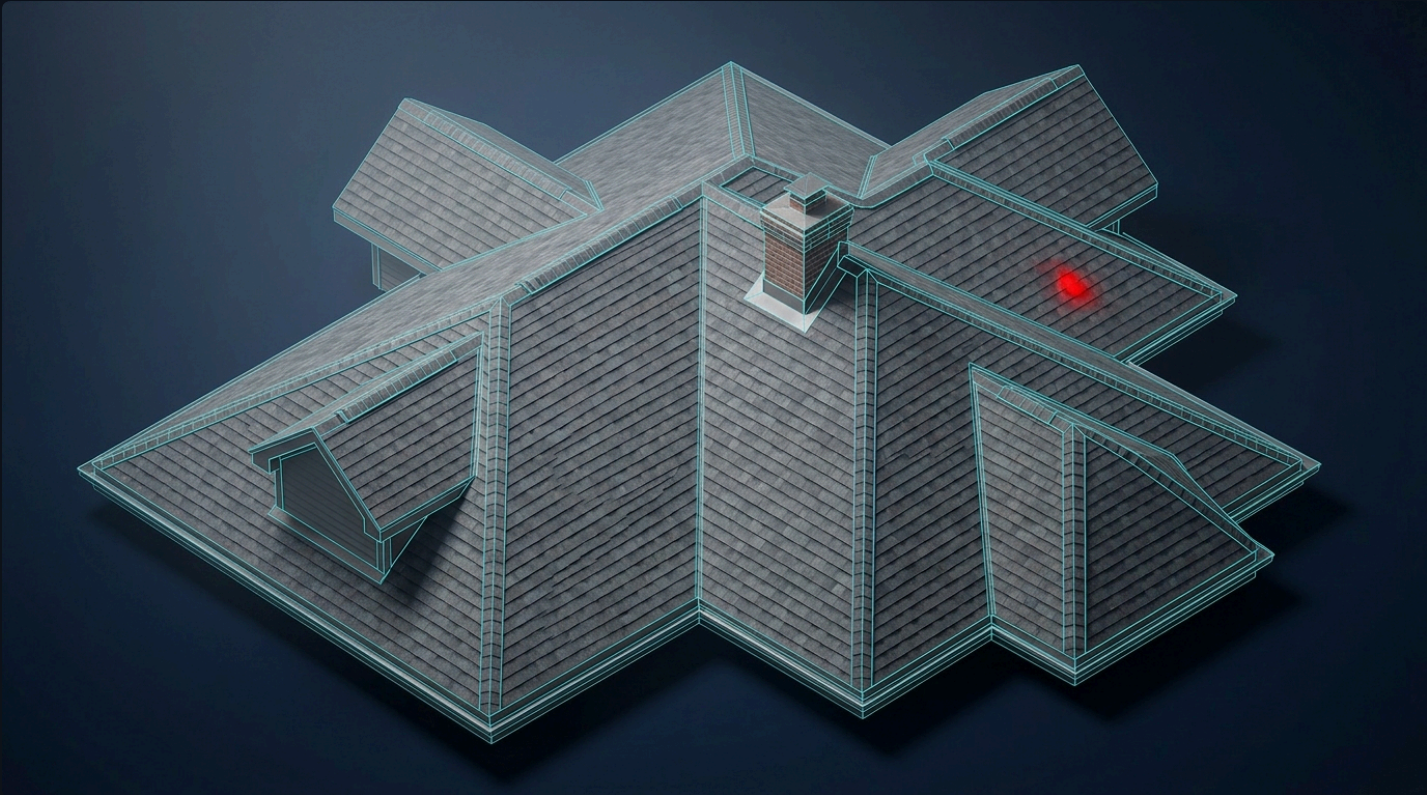
1	URGENT	Replace Front Door Weatherstripping	\$150/yr
2	HIGH	Remediate Blocked Attic Soffit Vents	\$110/yr
3	HIGH	Inspect and Seal 2nd-Story Window Headers	\$95/yr
4	MED	Add Attic Baffles for Cross-Ventilation	\$70/yr

THERMAL FORENSIC FINDINGS

Ridge Vent Performance	LOW FLOW	MED
Soffit Vents Open	8 / 22	HIGH
Front Door Seal Failure	SEVERE	SEVERE
2nd-Story Window Headers	WARM (+4.1°C)	HIGH

Sub-surface Anomaly Atlas

RADIOMETRIC OVERLAY • ROOF



CYAN: scan-vector overlay • ORANGE: thermal hotspot • Confidence threshold ≥ 85%

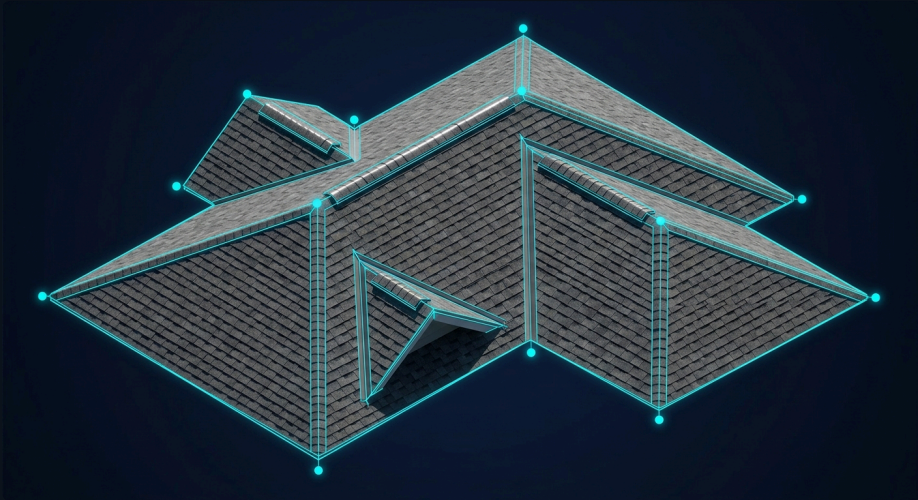
ANOMALY LEDGER

SEV	ID	TYPE	LOCATION	CONF	SF	DIAGNOSIS	EST
URGENT	AD-KY041-004	CDX Deck Rot	Facet F2 (Rear Hip)	92.3%	164.2	Trapped sub-surface moisture, multi-season propagation.	\$17,645.00
HIGH	AD-KY041-007	Soffit Blockage	North eave run	88.1%	0.0	Cellulose insulation blocking 14 of 22 soffit vents.	\$1,240.00
HIGH	AD-KY041-011	Window Header Leak	2nd story front bay	86.0%	0.0	Capillary water intrusion at flashing.	\$980.00



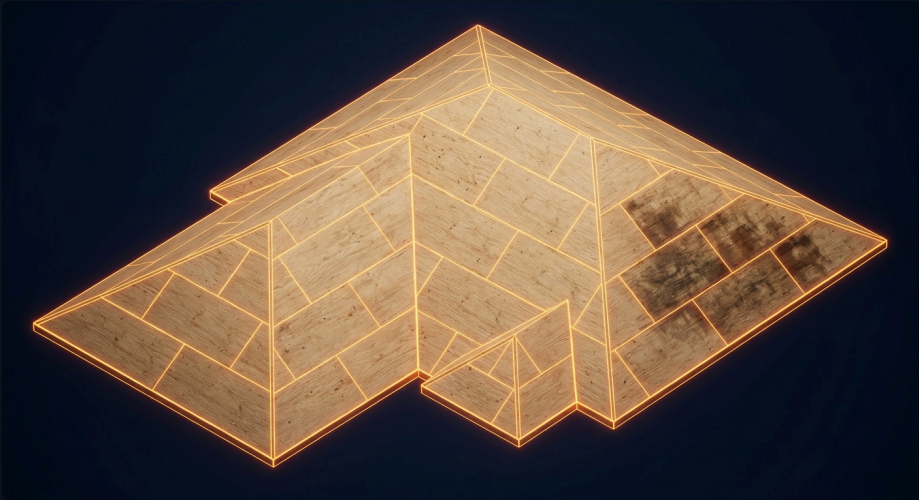
Layered CAD/BIM Twin

LAYER 01 • FINISHED SLATE



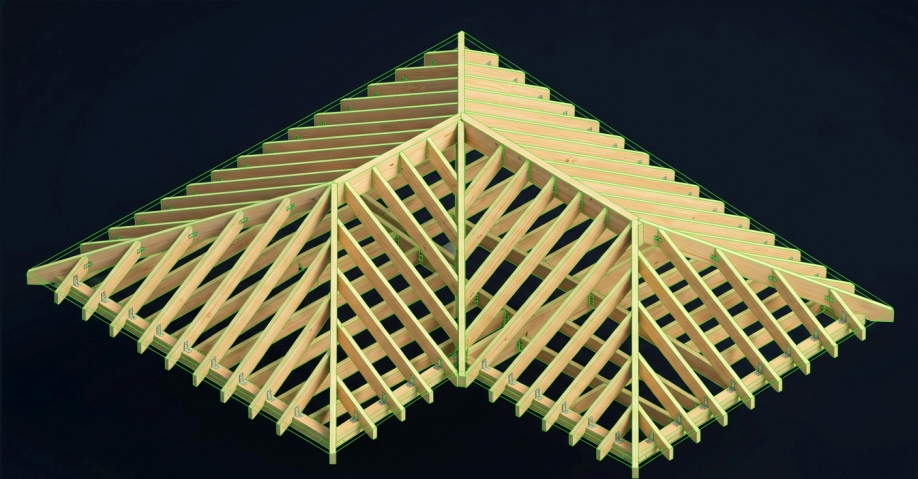
Squares 24.31 • Pitch 8/12

LAYER 02 • DECKING + UNDERLAYMENT



7/16" CDX • I&W shield at valleys/eaves • 2 existing shingle layer(s)

LAYER 03 • STRUCTURAL FRAMING



2x8 rafters @ 16" o.c. • LVL ridge • Hurricane ties **VERIFIED**

VALLEYS

148.7

LINEAR FT

GABLES

192.3

LINEAR FT

RIDGES

62.0

LINEAR FT

EAVES

188.4

LINEAR FT

Fenestration Schedule

ID	LOCATION	SHAPE	W	H	QTY	U-FACTOR	LEAK SEV
W1	Front Bay (1st)	Double-Hung	36"	60"	4	0.32	MED
W2	2nd Story Front	Double-Hung	32"	54"	5	0.32	HIGH
W3	Picture (Living)	Picture	72"	60"	1	0.28	LOW
W4	Bath / Kitchen	Awning	28"	32"	3	0.31	LOW
W5	Garage Service	Slider	48"	36"	1	0.34	MED

Exterior DoorSchedule

ID	LOCATION	SHAPE	W	H	QTY	WEATHERSTRIP
D1	Front Entry	Fiberglass 6-panel	36"	80"	1	SEVERE
D2	Rear / Patio	Sliding Glass	72"	80"	1	FAILED
D3	Garage to House	20-min Fire	32"	80"	1	WORN
D4	Garage Vehicle	Sectional Insulated	192"	84"	1	OK

Vertical Envelope + James Hardie Package

GROSS WALL SF

2840

PRE-SUBTRACTION

FENESTRATION

286

WINDOWS + DOORS

NET WALL SF

2554

SIDING TARGET

WALL HEIGHT

18.5'

EAVE-TO-GRADE

ELEVATION BREAKDOWN

ELEV.	GROSS SF	NET SF	MOISTURE
Front	780.0	642.0	14%
Rear	780.0	716.4	36%
Left	640.0	597.6	9%
Right	640.0	597.6	11%

SIDING PACKAGE • James Hardie

PROFILE: HardiePlank Lap • 8.25" exposure • COLOR: Iron Gray

ACCESSORY	QTY	UNIT	UNIT	LINE
HardiePlank Lap Siding	2554	sf	\$2.85	\$7,278.90
HardieTrim 4/4 (corners)	196	lf	\$4.40	\$862.40
HardieTrim Window/Door 4/4	312	lf	\$4.40	\$1,372.80
Starter Strip 1.25"	188	lf	\$1.90	\$357.20
J-Channel (Window/Door)	312	lf	\$1.45	\$452.40
Aluminum Soffit (vented)	540	sf	\$3.10	\$1,674.00
Aluminum Fascia 8"	188	lf	\$4.20	\$789.60
5" K-Style Aluminum Gutter	188	lf	\$7.40	\$1,391.20
3x4 Downspout w/ kick-out	6	ea	\$38.00	\$228.00
Tyvek HomeWrap 9' x 100' roll	3	rolls	\$175.00	\$525.00
Hardie 6d Stainless Nail (pail)	2	pails	\$92.00	\$184.00
Sealant • ColorPlus Touch-Up	6	tubes	\$14.50	\$87.00

SIDING SUBTOTAL\$15,202.50

Sub-SidingMoisture Atlas

VAPOR BARRIER STATUS

DEGRADED

BARRIER PRESENT • INFRARED-CONFIRMED

ZONES FLAGGED

4

THERMAL ANOMALY ZONES • WALL CAVITY

REMEDIATION TOTAL

\$3,120.00

ROLLED INTO MATERIALS TOTAL

THERMAL SATURATION ZONES • WALL CAVITY

ID	ELEV.	AREA SF	MOISTURE	DAMAGE PROB.	REMEDIATION	EST.
WMZ-01	Rear	64.0	38%	72%	Open wall • replace OSB sheathing + Tyvek	\$1,840.00
WMZ-02	Rear	32.0	28%	48%	Spot vapor-barrier patch + insulation dry-out	\$620.00
WMZ-03	Front	18.0	22%	34%	Re-flash bay window header • re-caulk	\$480.00
WMZ-04	Right	12.0	19%	26%	Replace exterior outlet gasket + spray-foam	\$180.00

Envelope **Air-Leak** Profile

BLOWER-DOOR ACH₅₀ (EST.)

8.4

TARGET ≤ 3.0 • LEAKY ENVELOPE

ANNUAL ENERGY LOSS

21,450_{kBTU}

SEASONAL LOSS • CONDITIONED ENVELOPE

ANNUAL DOLLAR LOSS

\$1,095.00_{/yr}

RECOVERABLE WITH REMEDIATION

LEAK-SOURCE LEDGER

SOURCE	LOCATION	BTU LOSS	\$/YR	RECOMMENDED FIX
Front Door Seal Failure	Front Entry D1	22%	\$241/yr	Replace weatherstripping + door sweep
2nd-Story Window Headers	Front W2 (×5)	18%	\$197/yr	Re-flash + foam-seal header cavities
Sliding Patio Door Track	Rear D2	14%	\$153/yr	Replace track gaskets + adjust roller
Attic Hatch / Pull-Down	2nd-floor hall	11%	\$120/yr	Install insulated hatch cover + weatherstripping
Rim-Joist Air Infiltration	Basement perimeter	13%	\$142/yr	Closed-cell spray foam rim joists (188 lf)
Garage-to-House Door	D3	8%	\$88/yr	New weatherstrip + auto-door-bottom
Recessed-Can Lights (un-rated)	Cathedral ceiling	14%	\$154/yr	Swap to IC-AT rated LED + air-seal

Bill of Materials

CATEGORY	MATERIAL	SKU	QTY	UNIT	UNIT COST	LINE TOTAL
Underlayment	#15 Felt Roll	SKU-FELT-15	16	rolls	\$49.00	\$784.00
Underlayment	Ice & Water Shield 36"	SKU-IW-75	4	rolls	\$110.00	\$440.00
Metal Flashing	Aluminum Drip Edge 10' Black	SKU-DRIP-10	17	10' sections	\$9.50	\$161.50
Metal Flashing	5x7 Aluminum Step Flashing	SKU-STEP-5F	80	ea	\$0.85	\$68.00
Metal Flashing	Aluminum Roll Flashing 16"	SKU-ROLL-ALU	1	50' roll	\$64.00	\$64.00
Shingles	Starter Strip (Three-Tab)	SKU-STARTER	3	bundles	\$34.00	\$102.00
Shingles	Hip & Ridge Cap (Three-Tab)	SKU-HIP-CAP	5	bundles	\$32.00	\$160.00
Shingles	Architectural Dimensional (Owens Corning Duration)	SKU-SHINGLE-V	74	bundles	\$33.00	\$2,442.00
Ventilation	Continuous Shingle-Over Ridge Vent 4'	SKU-RIDGE-V-L	12	4' sections	\$19.50	\$234.00
Ventilation	RV-28 Static Slant Back Vent (Black)	SKU-SLANT-BLK	4	ea	\$19.50	\$78.00
Fasteners	1" Plastic Cap Nails (7,500 ct)	SKU-CAP-1	1	box	\$19.50	\$19.50
Decking	7/16" CDX Plywood 4x8	SKU-CDX-716	14	sheets	\$38.00	\$532.00
MATERIALS TOTAL						\$23,407.50

Crew Economics

RATE BENCHMARK

NATIONAL AVG / SQ

\$285.00

REGIONAL AVG / SQ

\$248.00

REGION: KY / Mid-South • APPLIED: \$248.00/SQ

CREW PLAN

CREW SIZE

4

DAYS

2.5

LABOR TOTAL

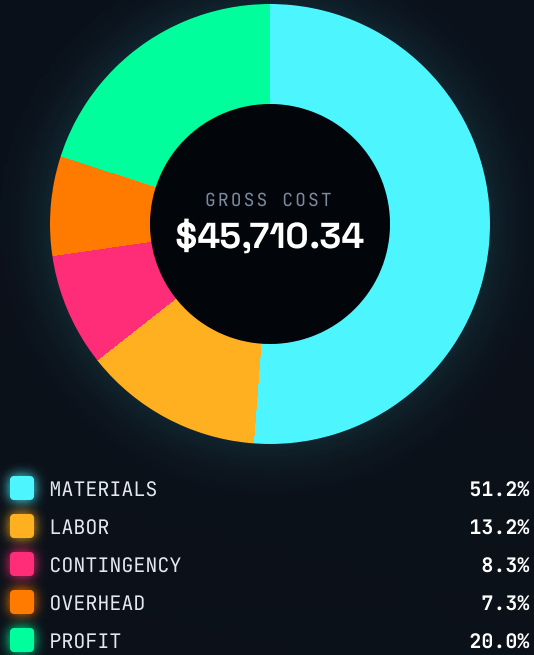
\$6,028.88

Crew Choreography

PHASE	CREW	DURATION	TIMELINE • 2.7 d
Mobilization & Site Setup	Supervision	0.25 d	
Tear-Off (existing 2 layers)	Roof Crew	0.50 d	
Decking Remediation (CDX rot)	Carpentry	0.50 d	
Underlayment + I&W Shield	Roof Crew	0.40 d	
Shingle Install (24.3 sq)	Roof Crew	0.70 d	
Wall Cavity Vapor Repair	Carpentry	1.00 d	
Hardie Siding Hang + Trim	Sheet-Metal	0.90 d	
Gutter + Downspout Install	Sheet-Metal	0.20 d	
Weatherstrip / Energy Sealing	Carpentry	0.10 d	
Cleanup + Final Inspection	Supervision	0.10 d	

Project Margin Overview

COST COMPOSITION



FINANCIAL SUMMARY

GROSS PROJECT REVENUE	\$45,710.34
TOTAL MATERIALS	\$23,407.50
TOTAL LABOR	\$6,028.88
CONTINGENCY (tear-off + reserve)	\$3,807.50
OVERHEAD (10%)	\$3,324.39
PROJECTED NET PROFIT (20%)	\$9,142.07

Project margin reasonable for forensic insurance restoration: 20 % net under 10/20 O&P caps. Retail equivalent would carry 28-32 % net.

Unforeseen Repair Reserve

ITEM	PROB	LOW	HIGH	NOTE
Additional CDX sheets (rot discovery)	65%	\$380.00	\$950.00	If hidden rot extends beyond Facet F2 once shingles are removed.
Chimney flashing rebuild	45%	\$425.00	\$880.00	Existing flashing shows oxidation; may not survive tear-off.
Fascia replacement (rear elevation)	30%	\$220.00	\$540.00	Sub-fascia rot suspected near AD-KY041-004 hotspot.
Bath fan vent re-route	20%	\$180.00	\$340.00	Currently dumping into attic – code violation if discovered.
RESERVE RANGE		\$1,205.00	\$2,710.00	

Full **Tear-Off + New Install** Required

WATER RETENTION PROBABILITY

88%

DEPTH EST 0.60" • MULTI-SEASON ENTRAPMENT

RATIONALE

Two existing shingle layers + 88% sub-surface moisture probability exceeds code re-roof limit. Full tear-off, decking remediation, and new-install is the only insurable path.

TEAR-OFF COST INCLUDED

\$1,850.00

Final Project Summary

TOTAL PROJECT COST

\$32,491.38

LOW • GRAND TOTAL

HIGH RANGE

\$33,996.38

HIGH • GRAND TOTAL

PROJECTED NET PROFIT

\$9,142.07

20% TARGET

URGENCY ITEMS

3

CRITICAL TASKS

AI-PRIORITIZED REPAIR PLAN

#	URGENCY	TASK	EST. ANNUAL SAVINGS
1	URGENT	Replace Front Door Weatherstripping	\$150/yr
2	HIGH	Remediate Blocked Attic Soffit Vents	\$110/yr
3	HIGH	Inspect and Seal 2nd-Story Window Headers	\$95/yr
4	MED	Add Attic Baffles for Cross-Ventilation	\$70/yr

CERTIFICATION

I certify the foregoing measurements, photographic evidence, thermal radiometry, and damage classifications were captured under STRATEX™ Phase-3 Expert Agent protocols with ground-truth accuracy of ±0.78 cm. This report is suitable for insurance carrier review.

INSPECTOR

STRATEX™ Forensic Division • A. Cross, RRO

KY-RR0-08821 • NRCIA-CRI-7741

REPORT ID

AD-KY041

ISSUED 2026-06-24

DISCLAIMER • This report is the proprietary work product of STRATEX™ Forensic Division. Findings are based on aerial drone reconnaissance, photogrammetric mesh reconstruction, and radiometric thermal analysis captured on the scan date. Concealed conditions discoverable only after tear-off are documented in the unforeseen-repair reserve. Reproduction or transmission without authorization is prohibited.